

Informatica® Data Replication
9.8.0

Release Guide

Informatica Data Replication Release Guide
9.8.0
December 2018

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Preface

The *Informatica Data Replication Release Guide* describes the cumulative changes for Informatica® Data Replication releases that are currently supported, beginning with version 9.6.2. Use this guide to get an overview of the changes in each release. This guide lists new features and enhancements, behavior changes, new and changed commands and parameters, and any newly supported versions of operating systems and databases. This guide also covers any removed functionality and deprecated versions of databases or operating systems.

Informatica Resources

Informatica Network

Informatica Network hosts Informatica Global Customer Support, the Informatica Knowledge Base, and other product resources. To access Informatica Network, visit https://network.informatica.com.

As a member, you can:

- Access all of your Informatica resources in one place.
- Search the Knowledge Base for product resources, including documentation, FAQs, and best practices.
- View product availability information.
- Review your support cases.
- Find your local Informatica User Group Network and collaborate with your peers.

Informatica Knowledge Base

Use the Informatica Knowledge Base to search Informatica Network for product resources such as documentation, how-to articles, best practices, and PAMs.

To access the Knowledge Base, visit https://kb.informatica.com. If you have questions, comments, or ideas about the Knowledge Base, contact the Informatica Knowledge Base team at KB_Feedback@informatica.com.

Informatica Documentation

To get the latest documentation for your product, browse the Informatica Knowledge Base at https://kb.informatica.com/_layouts/ProductDocumentation/Page/ProductDocumentSearch.aspx.

If you have questions, comments, or ideas about this documentation, contact the Informatica Documentation team through email at infa_documentation@informatica.com.

Informatica Product Availability Matrixes

Product Availability Matrixes (PAMs) indicate the versions of operating systems, databases, and other types of data sources and targets that a product release supports. If you are an Informatica Network member, you can access PAMs at https://network.informatica.com/community/informatica-network/product-availability-matrices.

Informatica Velocity

Informatica Velocity is a collection of tips and best practices developed by Informatica Professional Services. Developed from the real-world experience of hundreds of data management projects, Informatica Velocity represents the collective knowledge of our consultants who have worked with organizations from around the world to plan, develop, deploy, and maintain successful data management solutions.

If you are an Informatica Network member, you can access Informatica Velocity resources at http://velocity.informatica.com.

If you have questions, comments, or ideas about Informatica Velocity, contact Informatica Professional Services at ips@informatica.com.

Informatica Marketplace

The Informatica Marketplace is a forum where you can find solutions that augment, extend, or enhance your Informatica implementations. By leveraging any of the hundreds of solutions from Informatica developers and partners, you can improve your productivity and speed up time to implementation on your projects. You can access Informatica Marketplace at https://marketplace.informatica.com.

Informatica Global Customer Support

You can contact a Global Support Center by telephone or through Online Support on Informatica Network.

To find your local Informatica Global Customer Support telephone number, visit the Informatica website at the following link:
http://www.informatica.com/us/services-and-training/support-services/global-support-centers.

If you are an Informatica Network member, you can use Online Support at http://network.informatica.com.

CHAPTER 1

Version 9.8.0

This chapter includes the following topics:

- New Features and Enhancements, 9
- Parameter and Option Changes, 10
- Changes to Supported Operating Systems and Databases, 12

New Features and Enhancements

Informatica Data Replication 9.8.0 includes the following new features and enhancements:

Priorities for Oracle Archived Redo Log Locations

On the **Extract Range** tab in the Data Replication Console, you can specify multiple Oracle archive log destinations. The Console assigns priorities to the destinations based on their order in the list. The first destination in the list has the highest priority for use by the Oracle Extractor. If the Extractor cannot use the redo logs from the destination with the highest priority, it switches to the destination that has the next lower priority. You can change the order and priority of a destination by dragging and dropping it to a different position in the list.

If you specified multiple locations in a previous Data Replication version and then upgraded to 9.8.0, the Data Replication Console assigns priorities to the locations and lists the locations in order of priority on the **Extract Range** tab. If custom priorities with gaps were previously recorded in the config.db, these priorities are retained during the upgrade.

Support for MemSQL Targets

Data Replication 9.8.0 can now replicate data to MemSQL targets.

Reading Data from Redo Logs in Oracle RMAN Backup Sets

Data Replication can now extract change data from archived redo logs in backup sets that are created by the Recovery Manager (RMAN) utility.

To process redo logs from backup sets, set the extract.oracle.bsc.enable runtime parameter to 1.

For more information, see the *Informatica Data Replication User Guide*.

Support for Microsoft SQL Server Sources on Linux

Data Replication 9.8.0 can replicate data from Microsoft SQL Server 2017 sources on Linux.

Support for Microsoft SQL Server Compressed Backups of Transaction Logs

Data Replication can now capture data from Microsoft SQL Server compressed backups of transaction logs.

Parameter and Option Changes

Informatica Data Replication 9.8.0 introduces changes to the following parameters and options:

- Advanced runtime parameters that you configure on the **Runtime Settings** tab > **Advanced Settings** view in the Data Replication Console
- Server Manager advanced properties that you configure on the **Advanced Settings** tab of the **Properties** window in the Data Replication Console

Advanced Properties for the Server Manager

The following table describes changes to Server Manager advanced properties in Data Replication 9.8.0:

Property	New, Changed, or Deprecated	Description
EnforcePasswordHistory	New	Specifies the number of unique new passwords that users must define for their user accounts before they can use a previous password again. Default: 1
SessionExpirationTimeout	Changed	The time that is specified in the Server Manager SessionExpirationTimeout advanced property is now reset by console user activity as well as by network activity.
SqliteStatementsTracing	New	Indicates whether the Server Manager prints tracing messages for SQLite statements in the Data Replication execution logs. Valid values are: - **0**. Disable tracing for SQLite statements in the Data Replication execution logs. - **1**. Enable tracing for SQLite statements in the Data Replication execution logs. Default: 0

For more information, see the *Data Replication User Guide*.

Advanced Runtime Parameters

The following table describes changes to advanced runtime parameters in Data Replication 9.8.0:

Parameter	New, Changed, or Deprecated	Description
extractor.mssql.directory_for_decompressed_backup_logs	New	Specifies the name of the directory to which the Microsoft SQL Server Extractor decompresses compressed backup transaction logs before processing them. Default value: *DataReplication_installation*/tmp
extract.oracle.bsc.enable	New	Indicates whether the Oracle Extractor reads archived redo logs in backup sets that were created by the RMAN utility. Valid values are: - **0**. Do not process redo logs from backup sets. - **1**. Process redo logs from backup sets. Default value: 0
extract.oracle.bsc.number_of_threads_to_build_indexes	New	Specifies the number of threads that the Oracle Extractor can use for indexing archived redo logs in backup set containers. For this parameter to be used, the extract.oracle.bsc.enable parameter must be set to 1. Default value: 4 **Tip:** Increase the number of threads if the hard disk system is not fully loaded when the indexing of redo logs in backup set containers occurs. Decrease the number if the hard disk system usage reaches 100 percent.
extract.oracle.double_verification	Changed	Supports the following new valid values: - **2**. Verify the data integrity of all online redo logs. - **3**. Verify the data integrity of all online and archived redo logs.
extract.oracle.rds.fetch_archive_log_list_attempts	New	Specifies the maximum number of times that the Oracle Extractor retries a request for the list of archived redo log files from the Amazon RDS for Oracle instance. The Extractor retries the request at the interval that is specified in the extract.oracle.rds.fetch_archive_log_list_timeout runtime parameter. Default value: 5
extract.oracle.rds.fetch_archive_log_list_timeout	New	Specifies the interval, in seconds, that the Oracle Extractor waits after a request to get the list of archived redo log files from the Amazon RDS for Oracle instance fails before requesting the list again. Default value: 1

Parameter	New, Changed, or Deprecated	Description
extract.oracle.receive_timeout	New	Specifies the value that the Oracle Extractor uses for the Oracle SQLNET.RECV_TIMEOUT profile parameter if this parameter is not specified in the sqlnet.ora configuration file. The SQLNET.RECV_TIMEOUT parameter specifies the time, in seconds, for a client to wait for response data from the database server after establishing a connection. Default value: 600
extract.oracle.send_timeout	New	Specifies the value that the Oracle Extractor uses for the Oracle SQLNET.SEND_TIMEOUT profile parameter if this parameter is not specified in the sqlnet.ora configuration file. The SQLNET.SEND_TIMEOUT parameter specifies the time, in seconds, for a client to complete send operations to the database server after establishing a connection. Default value: 600

Changes to Supported Operating Systems and Databases

Data Replication 9.8.0 introduces the following changes to supported versions of operating systems and databases:

Replication Sources

The following table describes changes to the replication sources that Data Replication supports:

Source	Versions	New or Deprecated
DB2 for Linux, UNIX, and Windows	- 9.7	Deprecated
Microsoft SQL Server[1]	- 2017 on Linux	New
Microsoft SQL Server	- 2008 R2	Deprecated
MySQL	- 5.6	Deprecated
Oracle	- 18*c*	New

1. Data Replication 9.8.0 adds support for Microsoft SQL Server 2017 sources on Linux. You can now run the Extractor on the same Linux system as the Microsoft SQL Server source database to capture data from online transaction logs.

Replication Targets

The following table describes changes to the replication targets that Data Replication supports:

Target	Versions	New or Deprecated
Apache Kafka	- 2.0 - 1.1	New
Apache Kafka	- 0.10 - 0.8.2.2	Deprecated
DB2 for Linux, UNIX, and Windows	- 9.7	Deprecated
HortonWorks	- 2.3 - 2.2	Deprecated
Greenplum	- 4.2 - 4.1	Deprecated
MemSQL	- 6.0	New
MySQL	- 8.0	New
MySQL	- 5.6	Deprecated
Microsoft SQL Server	- 2008 R2	Deprecated
Netezza	- 6.*x* - 5.*x*	Deprecated
Oracle	- 18*c*	New
Teradata	- 14.10	Deprecated
Vertica	- 6.1 - 6.0	Deprecated

CHAPTER 2

Version 9.7.0 HotFix 2

This chapter includes the following topics:

- New Features and Enhancements, 14
- Parameter and Option Changes, 15
- Command Changes, 16
- Changes to Supported Operating Systems and Databases, 17

New Features and Enhancements

Informatica Data Replication 9.7.0 HotFix 2 includes the following new features and enhancements:

Enhanced Security

Informatica Data Replication 9.7.0 HotFix 2 includes the following new security features and enhancements:

Secure Hashing of User Passwords

Prior to Data Replication 9.7.0 HotFix 2, the Server Manager used the MD5 hashing algorithm for user passwords that are stored in the Server Manager SQLite database.

Now, Data Replication uses the more secure SHA512 hashing algorithm for user passwords.

Protection Against SQL Injections

Data Replication now uses a secure method of building SQL queries to configuration SQLite database files. This method protects Data Replication configurations against SQL injection attacks.

Idle Server Manager Sessions

When you close the Data Replication Console, the Console now closes its TCP/IP session to the Server Manager.

Also, the Server Manager closes Server Manager TCP/IP sessions that have no network activity and disconnects the Data Replication Console after the time that is specified in the Server Manager SessionExpirationTimeout advanced property elapses.

Parameter and Option Changes

Informatica Data Replication 9.7.0 HotFix 2 introduces changes to the following types of parameters and options:

- Advanced runtime parameters, which you configure on the **Runtime Settings** tab > **Advanced Settings** view in the Data Replication Console.
- Server Manager advanced properties, which you configure on the **Advanced Settings** tab of the **Properties** window in the Data Replication Console.

Advanced Runtime Parameters

The following table describes a new advanced runtime parameter:

Parameter	New, Changed, or Deprecated	Description
global.unicode_normalization	New	Indicates whether Data Replication converts source Unicode character data to Normalization Form C (NFC) format by using the International Components for Unicode (ICU) library. Valid values are: - **0**. Do not normalize Unicode character data. - **1**. Normalize Unicode character data. Use this option for Netezza targets if the source character data includes decomposed characters. Default value: 0 **Important:** If you set this parameter to 1 and the source character set matches the target character set, Data Replication converts the source data to UTF-16 for Unicode normalization. Then, the normalized Unicode character data is converted back to the source format.

Advanced Properties for the Server Manager

The following table describes changes to Server Manager advanced properties in Data Replication 9.7.0 HotFix 2:

Property	New, Changed, or Deprecated	Description
FilterOutConnectionsByUser	New	When the Data Replication Console or the Server Manager CLI requests information about the database connections or replication configurations, determines whether the Server Manager sends information about all of the replication objects or only about those objects for which the current Data Replication user is designated as the owner. Valid values are: - **0**. The Server Manager provides information about all of the database connections and configurations. - **1**. The Server Manager provides information about the database connections and configurations for which the current Data Replication user is designated as the owner. Default: 1
SessionExpirationTimeout	New	Specifies the amount of time, in seconds, that must elapse before a Server Manager TCP/IP session with no network activity times out and disconnects the Data Replication Console. Valid values: 0 through 2147483647. A value of 0 disables this timeout and causes the session to not disconnect from the Console. Default: 300 seconds

For more information, see the *Data Replication User Guide*.

Command Changes

Informatica Data Replication 9.7.0 HotFix 2 introduces changes to a Server Manager CLI command.

Server Manager CLI Commands

The following table describes a change to a Server Manager CLI command::

Command	New, Changed, or Deprecated	Description
start task	Changed	Includes the new optional parameter -d to run a replication task with its dependencies.

For more information, see the *Data Replication Command Line Interface for the Server Manager*.

Changes to Supported Operating Systems and Databases

Data Replication 9.7.0 HotFix 2 introduces the following changes to supported versions of operating systems and databases:

Operating Systems

The following table describes changes to the operating system versions that Data Replication supports:

Operating System	Versions	New or Deprecated
AIX	- 7.2 TL2	New
Redhat Enterprise Linux	- 7.1 - 7.2 - 7.3 - 7.4 - 6.6 - 6.7 - 6.8 - 6.9	New
Windows	- Windows Server 2016 - Windows 10	New

Replication Sources

The following table describes changes to the replication sources that Data Replication supports:

Source	Versions	New or Deprecated
DB2 for Linux, UNIX, and Windows	- 11.1	New
Microsoft SQL Server	- 2017	New

Replication Targets

The following table describes changes to the replication targets that Data Replication supports:

Target	Versions	New or Deprecated
Microsoft SQL Server	- 2017	New
DB2 for Linux, UNIX, and Windows	- 11.1	New
Teradata	- 16.00 - 16.10 - 16.20	New
Vertica	- 8.*x* - 9.*x*	New
Greenplum	- 5.*x*	New

Target	Versions	New or Deprecated
Netezza	- 7.2	New
PostgreSQL	- 10.*x*	New
Apache Kafka	- 0.10 - 0.11 - 1.0	New
Cloudera	- CDH 5.8 - CDH 5.9 - CDH 5.10 - CDH 5.11 - CDH 5.12 - CDH 5.13 - CDH 5.14	New
HortonWorks	- 2.4 - 2.5 - 2.6	New

CHAPTER 3

Version 9.7.0 HotFix 1

This chapter includes the following topics:

- New Features and Enhancements, 19
- Changes to Supported Operating Systems and Databases, 19

New Features and Enhancements

Informatica Data Replication 9.7.0 HotFix 1 includes the following new features and enhancements:

Support for Oracle 12c R2 Sources

Data Replication 9.7.0 HotFix 1 can now replicate data from Oracle 12*c* R2 sources.

Changes to Supported Operating Systems and Databases

Data Replication 9.7.0 HotFix 1 introduces the following changes to supported versions of databases:

Replication Sources

The following table describes changes to the replication sources that Data Replication supports:

Source	Versions	New or Deprecated
Oracle	- 12*c* R2	New

CHAPTER 4

Version 9.7.0

This chapter includes the following topics:

- New Features and Enhancements, 20
- Behavior Changes, 25
- Parameter and Option Changes, 26
- Command Changes, 36
- Changes to Supported Operating Systems and Databases, 37

New Features and Enhancements

Informatica Data Replication 9.7.0 includes the following new features and enhancements:

Support for Transparent Data Encryption of Microsoft SQL Server Databases

Data Replication 9.7.0 can now extract and decrypt change data from Microsoft SQL Server source databases that use Transparent Data Encryption (TDE).

For more information, see the *Informatica Data Replication User Guide*.

Secure Connections to Greenplum Targets

Data Replication 9.7.0 can now use the TCP/IP protocol with SSL to securely connect to Greenplum targets.

For more information, see the *Informatica Data Replication User Guide*.

Reinstated Support for PostgreSQL Targets

The Data Replication 9.7.0 release reinstates support for the PostgreSQL targets that were temporarily not supported in version 9.6.1 and later.

The following table describes changes to datatype mappings from Microsoft SQL Server sources to PostgreSQL targets:

Microsoft SQL Server Source Datatype	PostgreSQL Target Datatype	Comments

VARCHAR(*n*)	VARCHAR(*n*)	Prior to Data Replication 9.7.0, the following PostgreSQL target datatype mappings were recommended for SQL Server source VARCHAR columns:
		- VARCHAR(*n*)
		Mapping condition:
		`n <= 255`
		- TEXT
		Mapping condition:
		`n > 255`

MONEY	NUMERIC(19,4)	Prior to Data Replication 9.7.0, the recommended datatype mapping for SQL Server source MONEY columns was PostgreSQL target MONEY columns.

Support for Microsoft SQL Server 2016 Sources

Data Replication 9.7.0 can now replicate data from Microsoft SQL Server 2016 sources except sources that use the Always Encrypted option.

Support for MySQL Sources

Data Replication 9.7.0 can now replicate change data from MySQL Enterprise 5.7.*x* sources to any supported target type.

With MySQL sources, the Extractor and InitialSync must run on a Linux or Windows operating system.

Important: If your *DataReplication_installation*/uiconf/ directory contains the default.cfg file, you must set the show_mysql_source_db_tab parameter in this file to true to enable support for MySQL sources.

Data Replication supports MySQL source databases that use the InnoDB storage engine only. Data Replication reads change events from binary log files that describe the DML and DDL changes to the database.

Data Replication can capture changes for source tablespaces that use InnoDB tablespace encryption and for InnoDB source tables that use Transparent Data Encryption (TDE).

The following Data Replication features are not supported for configurations with MySQL sources:

- Importing and exporting Start Point or Sync Point log coordinates
- Performing bidirectional replication
- Managing open transactions from the Data Replication Console or Server Manager CLI

For more information, see the "Sources - Preparation and Replication Considerations" chapter of the *Informatica Data Replication User Guide*.

Backing up Replication Configurations During Clean Operations

When you request a clean operation on a configuration, Data Replication 9.7.0 can now optionally back up the existing configuration SQLite database files and intermediate files before starting clean processing.

The following information is backed up:

- All of the SQLite database files that are associated with the replication configuration
- All of the intermediate files that are associated with the replication configuration

Data Replication stores the backups in the directory that is specified in the **Backups** field on the **Paths** tab of the Server Manager **Properties** dialog box. Data Replication creates a subdirectory for each clean operation using the following naming pattern: *<current_timestamp>_<configuration_name>*.

By default, Data Replication stores each backup for 30 days. Data Replication checks the backup subdirectories hourly to remove any subdirectory for which the retention period has expired. You can use the Server Manager TimeToStoreBackups advanced property to change the number of days that the Server Manager Main server stores the backups.

For more information, see the *Informatica Data Replication User Guide*.

Filtering Change Data Records in the Intermediate File Viewer by SCN Values

When you view intermediate files for a configuration in the Data Replication Console, you can now filter change records by minimum and maximum SCN values.

For more information, see the *Informatica Data Replication User Guide*.

Support for Apache Kafka Targets

Data Replication 9.7.0 can now replicate change data from any supported source type to an Apache Kafka messaging system.

The Kafka Applier uses an intermediate Java-based tool, called the *Informatica Queue Adapter*, to process and format change data from a source database into Kafka messages and then connect to and send the messages to a consumer application. The Kafka Applier sends a single stream of completed transactions, in commit order, to the Queue Adapter. Data Replication provides the required Queue Adapter library files in the *DataReplication_installation*/lib/queueAdapterLib subdirectory.

The Kafka messages are published and subscribed to based on Kafka topic names. You can use one topic for all messages or a separate topic for each source table. You must create the topics or configure Kafka to automatically create the topics.

The messages include most the same metadata fields that are in audit log tables. You can customize the names of these fields in the Data Replication Console.

For recovery after a failure, Kafka targets use a checkpoint file instead of a recovery table. Data Replication saves the sequence of the last successful change operation to the checkpoint file.

The following limitations and restrictions pertain to Kafka targets:

- Data Replication supports only Audit Apply mode for Kafka targets.
- Data Replication supports Kafka version 0.8.2.2 or later on Linux and Windows operating systems only.
- The Kafka client library files must match the version of Kafka to which Data Replication connects, and the files must be located in a directory that runs on the same machine as the Data Replication Applier

- You must use an Avro schema in nested or flat format to format the messages to be published to Kafka topics. The consumer application that subscribes to the topics must also have the Avro schema that is required to read the messages. You can generate the consumer schema from the Data Replication Console.
- The messages can use an Avro encoding type of JSON or binary.
- To attempt to guarantee that no messages are lost or duplicated on the Kafka target when a network outage occurs, Data Replication enables guaranteed delivery of messages to Kafka targets by default. Guaranteed delivery produces the highest data accuracy but might reduce performance. When guaranteed delivery is disabled, duplicate messages might appear on the Kafka target when you restart replication after a network outage.
- The following Data Replication features are not supported for Kafka targets.
  - InitialSync
  - DDL replication
  - Data Replication SQL expressions
  - Replication from a single source to multiple Kafka targets, bidirectional replication, and cascade replication

Before you create a replication configuration with a Kafka target, configure connectivity information in the Kafka producer.properties file on the system where the target Server Manager and Applier run.

Many new runtime parameters are available for Kafka targets and the Queue Adapter. These parameters begin with app.kafka.kafka_, apply.avro.avro_, or apply.qadapter.

For more information, see the *Informatica Data Replication User Guide*.

Collecting Diagnostic Data for Troubleshooting

You can use Data Replication to collect diagnostic data for troubleshooting the Data Replication Console and Data Replication tasks. Typically, you collect diagnostic data at the request of Informatica Global Customer Support.

Data Replication can collect the following information:

- Data Replication SQLite database files
- Task execution logs
- Server Manager logs
- Intermediate files
- Data Replication Console logs

Note: You can choose which files to include in the diagnostic data collection based on various criteria, such as the configuration and specified time period.

To collect diagnostic data about Data Replication tasks, Data Replication creates a Collect Diagnostic Data task and starts the task with the data collection settings that you define in the Data Replication Console.

Important: Only the idradmin user can create Collect Diagnostic Data tasks. However, if you do not delete a Collect Diagnostic Data task after collecting diagnostic data, any user can start the task again and collect the same set of diagnostic data.

After Data Replication collects the requested diagnostic data, the Server Manager Main server generates the diag.zip file in the *DataReplication_installation*/logs/*date*/*run_id* subdirectory.

Restriction: The Collect Diagnostic Data task cannot collect a requested file from a Data Replication installation on Windows if the file path is longer than the Windows limit of 260 characters.

To collect diagnostic data about the local Data Replication Console installation, you do not have to connect to a Server Manager. Any user can create a zip file with the Data Replication Console log files.

For more information, see the *Informatica Data Replication User Guide*.

Support for DB2 for Linux, UNIX, and Windows Multidimensional Clustering Tables

Data Replication can now extract change data from DB2 for Linux, UNIX, and Windows multidimensional clustering (MDC) tables.

For more information, see the *Informatica Data Replication User Guide*.

Improved Recovery Processing

Prior to Data Replication 9.7.0, you generated a recovery table in the Data Replication Console when saving a replication configuration. If an Applier or database failure occurred and you used Audit Apply mode with the direct load method or used Merge Apply mode, recovery might result in duplicate data or not occur.

Starting with Data Replication 9.7.0, the Applier transparently generates a recovery table on the target when it runs for the first time. The schema of the recovery table has been updated. Recovery processing is now more resilient in Audit Apply mode.

For more information, see the *Informatica Data Replication User Guide*.

Extractor Start Point Management During Clean Operations

Prior to Data Replication 9.7.0, after a clean operation, the Extractor Start Point was not updated. Consequently, extraction processing began back at the default Extractor Start Point value, which caused some changes to be extracted again.

In Data Replication 9.7.0, after a clean operation, the Extractor Start Point is set to the last SCN, LSN, or log coordinates value that the Applier processed for the configuration. If an open transaction exists for the mapped source tables, the Extractor Start Point is set to the SCN, LSN, or log coordinates value for the BEGIN TRANSACTION operation.

For more information, see the *Informatica Data Replication User Guide*.

Support for Named Source and Target Connections Only

Data Replication 9.7.0 now supports only named source or target connections. You can no longer enter connection details without a unique name for a configuration.

In the Data Replication Console, you can create named connections either from the **Server Manager** tab > **Connections** view before you create replication configurations or from the **Source Database** or **Target Database** tab when you define a configuration. During configuration creation, you can either select a source or target connection that is already defined or create a new connection definition.

All connection definitions are listed on the **Server Manager** tab > **Connections** view by database type and are available for inclusion in multiple configurations. You can edit a connection only from the **Connections** view.

When you deploy a configuration or add a target on the **Routing** tab, you must select a predefined named connection. When you import a configuration, if you do not specify a named connection for the source or target connection, the Data Replication Console generates a connection definition with a unique name, based on the information in the imported configuration file.

If you are upgrading from a previous version of Data Replication and have configurations that use un-named connections, Informatica recommends that you save these connections under unique names before upgrading the Data Replication installation. If you do not perform this step, after the upgrade, the Server Manager will automatically create separate named connections for each configuration, even if multiple configurations use the same connection details. The names of these generated connections will have the format <*configuration_name*>_source_connection or <*configuration_name*>_target_connection. If a connection with the generated name already exists, the Server Manager adds the number 1 to the end of the name. If the connection name is still not unique, the Server Manager increments the number by 1 until the name is unique.

Also, the Server Manager CLI now includes the following new commands for creating and managing connections:

- add connection
- delete connection
- edit connection

For more information, see the "Creating and Managing Connections," "Creating Replication Configurations," and "Managing Replication Configurations" chapters in the *Informatica Data Replication User Guide*. Also see the "Command Reference" chapter in the *Informatica Data Replication Command Line Interface for the Server Manager*.

Support for Amazon Relational Database Service (RDS) for Oracle

Data Replication can now extract change data from Amazon Relational Database Service (RDS) for Oracle deployments of Oracle databases in the cloud.

To process redo logs from an Amazon RDS for Oracle instance, set the extract.oracle.rds_read_enable runtime parameter to 1.

Important: For Amazon RDS for Oracle instances, the Extractor processes only the redo logs that are located in the following directories:

- ARCHIVELOG_DIR for archived redo log files
- ONLINELOG_DIR for online redo log files

For more information, see the *Informatica Data Replication User Guide*.

Support for Oracle RESETLOGS Operations

Previously, Data Replication did not support database incarnations. The Extractor assumed that only a single log incarnation of the source database existed and processed transaction logs in the order of the log sequence numbers. When a user performed RESETLOGS and Oracle restarted transaction log numbering, the Extractor processed all transaction logs from both incarnations and ended with an error.

Data Replication can now continue extracting change data from the redo logs across the RESETLOGS boundary.

For more information, see the *Informatica Data Replication User Guide*.

Behavior Changes

Informatica Data Replication 9.7.0 introduces the following behavior changes.

Enabling Supplemental Logging for Source Tables with Unique Indexes That Have Only Nullable Columns

Prior to Data Replication 9.7.0, when you saved a configuration in the Data Replication Console, the Console checked whether supplemental logging was enabled for columns in a primary key or unique index that included at least one not-nullable column. If supplemental logging was not enabled for at least one of the columns that was checked, the Console prompted you to enable supplemental logging. The Console did not check whether supplemental logging was enabled if the primary key or unique index was composed of only nullable columns, primarily to prevent replication problems that could occur for Oracle source tables when all of the index columns contain a null value.

Note: Oracle allows a table to have multiple rows in which all unique index columns contain a null value. However, other source types do not allow this situation.

In Data Replication 9.7.0, if a table of any source type has a unique index that is composed of only nullable columns, the Console checks whether supplemental logging is enabled for the unique index columns and prompts you to enable supplemental logging if one more index columns are not logged. With supplemental logging enabled on the index columns, the Applier can build a WHERE clause to replicate Updates and Deletes. However, for Oracle sources, the Applier might then apply Update or Delete operations to an incorrect row if the source table has multiple rows in which all of the unique index columns contain a null value. To prevent this problem, *before* you run replication tasks for the configuration, verify that the table does not contain multiple rows in which all of the unique index columns contain a null. If necessary, you can add the NOT NULL constraint on an index column or ensure that the database user application does not generate rows in which all of the index columns can contain a null.

Parameter and Option Changes

Informatica Data Replication 9.7.0 introduces changes to the following parameters and options:

- Advanced runtime parameters that you configure on the **Runtime Settings** tab > **Advanced Settings** view in the Data Replication Console.
- Server Manager advanced properties that you configure on the **Advanced Settings** tab of the **Properties** window in the Data Replication Console.
- Command line parameters for the Extractor.

Advanced Runtime Parameters

The following table describes changes to advanced runtime parameters in Data Replication 9.7.0:

Advanced Runtime Parameter	New, Changed, or Deprecated	Description
apply.avro.avro_before_image_prefix	New	Specifies a prefix for the names of Avro message fields that contain the before image data for an Update operation. Valid values are the digits 0-9, Latin letters A-Z and a-z, and the underscore (_) character. If you specify a value for this parameter and the apply.avro.avro_before_image_suffix parameter, the Avro field name uses the following format: *<prefix><source_column_name><suffix>*. **Note:** If you do not specify either a prefix or suffix, the Avro before-image field name could duplicate the source column name. Informatica recommends that you enter a value in at least one of these parameters. For this parameter to be used, the apply.avro.avro_include_before_image parameter must be set to 1. Default value: Null
apply.avro.avro_before_image_suffix	New	Specifies a suffix for the names of Avro message fields that contain the before image data for an Update operation. Valid values are the digits 0-9, Latin letters A-Z and a-z, and the underscore (_) character. If you specify a value for this parameter and the apply.avro.avro_before_image_prefix parameter, the Avro field name uses the following format: *<prefix><source_column_name><suffix>*. **Note:** If you do not specify either a suffix or prefix, the Avro before-image field name could duplicate the source column name. Informatica recommends that you enter a value in at least one of these parameters. For this parameter to be used, the apply.avro.avro_include_before_image parameter must be set to 1. Default value: _OLD
apply.avro.avro_include_before_image	New	Indicates whether Data Replication creates Avro message fields that contain the before image of columns when replicating an Update operation to a Kafka target. Valid values are: - **0**. Do not create fields that contain the before image of columns. - **1**. Create fields that contain the before image of columns. Default value: 1

Advanced Runtime Parameter	New, Changed, or Deprecated	Description
apply.avro.avro_include_is_present	New	Specifies whether Data Replication creates Avro message fields that contain a Boolean value indicating one of the following reasons for the occurrence of a null value in an Avro message field: - The corresponding source column value is also null (null is present). - The corresponding source column has a value that is not null (null is not present). However, Data Replication does not capture the value because it did not change during the current replication cycle. Valid values are: - **0**. Do not create fields that contain a Boolean value indicating whether the null value is present or not present in the source data. - **1**. Create fields that contain a Boolean value indicating whether the null value is present or not present in the source data. Default value: 1
apply.avro.avro_is_present_prefix	New	Specifies a prefix for the names of Avro message fields that contain an indicator that a null value in the messages is also present in the corresponding source data. Valid values are the digits 0-9, Latin letters A-Z and a-z, and the underscore (_) character. You can also specify a suffix in the apply.avro.avro_is_present_suffix parameter. If you specify both a prefix and suffix, the Avro field name uses the following format: *<prefix><source_column_name><suffix>*. **Note:** If you do not specify a prefix or suffix, the Avro field name could duplicate the source column name. Informatica recommends that you enter a value in at least one of these parameters. For this parameter to be used, the apply.avro.avro_include_is_present parameter must be set to 1. Default value: Null

Advanced Runtime Parameter	New, Changed, or Deprecated	Description
apply.avro.avro_is_present_suffix	New	Specifies a suffix for the names of Avro message fields that contain an indicator that a null value in the messages is also present in the corresponding source data. Valid values are the digits 0-9, Latin letters A-Z and a-z, and the underscore (_) character. You can also specify a prefix in the apply.avro.avro_is_present_prefix parameter. If you specify both a suffix and prefix, the field name uses the following format: *<prefix><source_column_name><suffix>*. **Note:** If you do not specify a suffix or prefix, the Avro field name could duplicate the source column name. Informatica recommends that you enter a value in at least one of these parameters. For this parameter to be used, the apply.avro.avro_include_is_present parameter must be set to 1. Default value: _PRESENT
apply.avro.avro_metadata_prefix	New	Specifies a prefix for the names of any metadata fields that are added to an Avro schema to distinguish these metadata fields from source columns. You can also specify a suffix by using the apply.avro.avro_metadata_suffix parameter. If you specify both a prefix and suffix, the names of the metadata fields use the following format: *<prefix><metadata_field_name><suffix>*. Default value: Null
apply.avro.avro_metadata_suffix	New	Specifies a suffix for the names of any metadata fields that are added to an Avro schema to distinguish these metadata fields from source columns. You can also specify a prefix by using the apply.avro.avro_metadata_prefix parameter. If you specify both a suffix and prefix, the names of the metadata fields use the following format:*<prefix><metadata_field_name><suffix>*. Default value: Null
apply.avro.avro_name_replacement_value	New	Specifies a character that will replace any unsupported character in source column or table names when Data Replication generates an Avro field name for Kafka messages. Avro field names do not allow spaces or special characters. If this parameter is not specified, Data Replication will not replace unsupported Avro characters from source column or table names. For example, if the parameter is set to the underscore (_) character and a source column is named `column#name with space`, the corresponding Avro field name is `column_name_with_space`. Default value: Underscore (_) character

Advanced Runtime Parameter	New, Changed, or Deprecated	Description
apply.avro.avro_schema_cache_directory	New	Specifies the directory where the Avro formatting code stores Avro schema objects that are used to produce Avro messages for each mapped source table. Default value: *DataReplication_installation*/overflow
apply.avro.avro_schema_cache_memory_size	New	Specifies the maximum number of Avro schema files in a single Least Recently Used (LRU) cache instance to keep in memory. Data Replication writes an Avro schema file for each mapped source table in a configuration to the directory specified in the apply.avro.avro_schema_cache_directory parameter. When the number of Avro schema files in the cache instance exceeds the limit specified in this parameter, the excess files are removed from memory. Default value: 1000
apply.kafka.kafka_checkpoint_file_directory	New	Specifies the directory to which Data Replication writes checkpoint files when you apply data to a Kafka target. The directory must be on the system where the Applier runs. Default value: *DataReplication_installation*/checkpoints
apply.kafka.kafka_file_checkpoint_file_name	New	Specifies the name of the checkpoint file that Data Replication creates when you apply data to a Kafka target. When the value of this parameter is DEFAULT, each checkpoint file name matches the corresponding configuration name. Default value: DEFAULT
apply.kafka.kafka_client_libraries_directory	New	Specifies the directory that Data Replication uses to access Kafka client libraries. Copy the files from the *kafka_version*/libs directory to the default Data Replication directory or enter the path to another directory. The library files must match the version of Kafka to which Data Replication connects, and the files must be located in a directory that runs on the same machine as the Data Replication Applier. Default value: *DataReplication_installation*/lib/queueAdapterLib.

Advanced Runtime Parameter	New, Changed, or Deprecated	Description
apply.kafka.kafka_message_key	New	Specifies an optional message key that Data Replication uses to assign a topic partition to the Avro messages sent to a Kafka target. All messages that use the same topic and message key are assigned to the same partition in the topic. If you specify the USE_TABLE_NAME, Data Replication uses the name of each mapped source table as the key name in the messages. Default value: Null
apply.kafka.kafka_producer_guarantee_delivery	New	Specifies whether or not to enable guaranteed delivery of messages to a Kafka target. When guaranteed delivery is enabled, the Queue Adapter attempts to guarantee that no messages are lost or duplicated on the target if a network outage occurs. Guaranteed delivery provides for the highest level of data integrity on the target but might degrade performance. Valid values are: - **0**. Disable guaranteed delivery. The Applier writes a checkpoint to the checkpoint file after each Commit operation, or after the timeout period that is specified in the apply.qadapter.checkpoint_request_timeout runtime parameter elapses. If you restart apply processing after an outage, duplicate or missing messages might occur on the target. - **1**. Enable guaranteed delivery. The Applier writes a checkpoint to the checkpoint file after each message is successfully sent to the target topic. All messages are sent synchronously if the apply.kafka.kafka_producer_send_synchronously runtime parameter is set to 1. If you restart apply processing after an outage, no duplicate or missing messages should occur on the target. Default value: 1
apply.kafka.kafka_producer_max_retry_attempts	New	Specifies the maximum number of times that the Queue Adapter Connector will retry sending a message that produced an error. Default value: 5
apply.kafka.kafka_producer_partition_ID	New	Specifies the topic partition ID that Data Replication will include in each message passed to Kafka. A value of -1 causes Data Replication to send a null partition ID to Kafka. Default value: -1

Advanced Runtime Parameter	New, Changed, or Deprecated	Description
apply.kafka.kafka_producer_send_synchronously	New	Controls whether to enable synchronous delivery of messages to a Kafka target. When synchronous delivery is enabled, Kafka must acknowledge each message as received before Data Replication will send the next message. In this mode, Kafka is unlikely to receive duplicate messages, but performance might be slower. Valid values are: - **0**. Disable synchronous delivery. The Applier sends messages asynchronously to the target. - **1**. Enable synchronous delivery. Default value: 0 **Note:** With either option, the Queue Adapter maintains the order of changes streamed from the source.
apply.kafka.kafka_table_name_include_schema	New	When USE_TABLE_NAME is specified in the Data Replication Console or in the apply.kafka.kafka_message_key parameter to use the mapped source table name as the Kafka topic name or message key name, you can use this parameter to also include the source table schema name in the topic and key names. Valid values are: - **0**. Do not include the source table schema name in the topic and key names. - **1**. Include the source table schema name in the topic and key names. With this setting, a Kafka topic name will have the following format: *schema_name*.*source_table_name* Default value: 0
apply.kafka.kafka_table_name_prefix	New	When USE_TABLE_NAME is specified in the Data Replication Console or in the apply.kafka.kafka_message_key parameter to use the mapped source table name as the Kafka topic name or message key name, you can use this parameter to also include a prefix in the topic and key names. For example, if you specify a prefix in this parameter and set the apply.kafka.kafka_table_name_include_schema parameter to 1, a Kafka topic name will have the following format: *prefix_schema_name*.*source_table_name*. Default value: Null
apply.qadapter.cache_overflow_directory	New	Specifies the directory that Data Replication uses to write Least Recently Used (LRU) cache overflow files from the Queue Adapter. Default value: *DataReplication_installation*/overflow.

Advanced Runtime Parameter	New, Changed, or Deprecated	Description
apply.qadapter.cache_memory_size	New	For the Queue Adapter, specifies the maximum number of items in a single Least Recently Used (LRU) cache instance to keep in memory. When the number of cached items exceeds this limit, the overflow items are stored in the overflow directory specified in the apply.qadapter.cache_overflow_directory parameter. Default value: 1000
apply.qadapter.checkpoint_request_timeout	New	Specifies the time interval (in milliseconds) that the Applier waits for the Queue Adapter to validate that all data was successfully sent to a Kafka target. The Applier typically requests this checkpoint information at the end of each Apply cycle before the intermediate files are deleted. If the Queue Adapter does not validate that all data was sent to a Kafka target before this time elapses, the Queue Adapter reports a timeout error. The Applier then checks for error conditions, retries the request, or fails. Use this parameter and the apply.qadapter.checkpoint_retries parameter to provide enough time for the Applier to write all data to a Kafka target so that Data Replication can release the intermediate files for the Apply cycle. **Note:** If Data Replication persistently issues an IDR-070218 error message and reports a syncKafkaEnvironment failure, try adjusting the apply.qadapter.checkpoint_request_timeout and apply.qadapter.checkpoint_retries parameter values. Default value: 60000 (60 seconds)
apply.qadapter.checkpoint_retries	New	After a timeout error, specifies the maximum number of consecutive times that the Applier retries a request for the Queue Adapter to validate that all data was successfully sent to a Kafka target. If the Queue Adapter does not validate that all data was sent to a Kafka target after the maximum number of retries, the Applier ends with an error. For example, if the value of the apply.qadapter.checkpoint_request_timeout parameter is 60000 (60 seconds) and the value of this parameter is 6, 6 minutes must expire before the Applier stops waiting for all data to be written to a Kafka target and ends with an error. **Note:** If Data Replication persistently issues an IDR-070218 error message and reports a syncKafkaEnvironment failure, try adjusting the apply.qadapter.checkpoint_request_timeout and apply.qadapter.checkpoint_retries parameter values. Default value: 5

Advanced Runtime Parameter	New, Changed, or Deprecated	Description
apply.qadapter.create_missing_directories	New	When the directories that are specified in the apply.kafka.kafka_checkpoint_file_directory and apply.qadapter.cache_overflow_directory parameters do not exist, determines whether to create these directories or report an access error. Valid values are: - **0**. Report an access error, and do not create the directories. - **1**. Create the directories. Default value: 1
apply.qadapter.properties_file	New	Specifies a file that contains additional Queue Adapter properties for use by Informatica Global Customer Support or other Informatica personnel. The properties in this file are used to change or augment default Queue Adapter behavior with Data Replication. Edit the value of this parameter only at the direction of Informatica Global Customer Support.
apply.threshold_time_for_open_transaction	New	Specifies the number of minutes that a transaction must be open before the Applier logs a warning message about the open transaction. The Applier compares this threshold time to the time the transaction has been open, which is calculated as the difference between the time of the last record from the open transaction and the time of the last processed record from any transaction. Default value: 60 minutes
extract.mssql.ignore_inactive_backup_locations	New	If you add shared backup log locations on the **Extract Range** tab, indicates whether the Extractor ends with an error when a location becomes temporarily unavailable. Options are: - **0**. The Extractor ends with an error if it cannot access a backup log location. - **1**. The Extractor continues processing even if it cannot access a backup log location. Use this option for Microsoft SQL Server sources that use Always On Availability Groups. Default value: 0
extract.oracle.rds_read_enable	New	Controls whether the Extractor can extract change data from online and archived redo logs for Amazon Relational Database Service (RDS) for Oracle instances in the cloud. Options are: - **0**. The Extractor cannot process redo logs from Amazon RDS for Oracle instances. - **1**. The Extractor can process redo logs from Amazon RDS for Oracle instances. Default value: 0

Advanced Runtime Parameter	New, Changed, or Deprecated	Description
extract.oracle.skip_lob_data_for_table	New	Specifies the name of an Oracle source table for which you want the Extractor to replace LOB data with null values in the intermediate files. Use this parameter for a table that includes SecureFile LOB columns that use unsupported options, such as NOLOGGING or FILESYSTEM_LIKE_LOGGING. Otherwise, the Extractor ends with an error. Default value: Empty
extract.oracle.threads_online_at_start	New	Specifies a comma-separated list of the Oracle redo threads that are online at the SCN that corresponds to the Start Point value for the Extractor. The Extractor starts processing change data records from Oracle redo threads that are in this list. The Data Replication Console populates this list of online redo threads. Informatica recommends that you do not change the populated values for this parameter.
global.nz_pg_empty_strings_as_null	Changed	This parameter now applies to the PostgreSQL target type in addition to the following previously supported target types: - Greenplum - Vertica - Netezza
initial.oracle_clob_text_buffer_size	Changed	The default value for Teradata targets changed from 0 to the global.lob_truncation_size runtime parameter value.

Advanced Properties for the Server Manager

The following table describes changes to Server Manager advanced properties in Data Replication 9.7.0:

Property	New, Changed, or Deprecated	Description
SQLiteBusyTimeout	New	Specifies the number of milliseconds that the Server Manager or a replication task can wait to open the configuration SQLite database to run a SQL statement when the database is locked by another task. When the specified time elapses, the request to the SQLite database fails. Valid values: 0 through 2147483647 Default value: 30000
TimeToStoreBackups	New	Specifies the number of days that the Server Manager Main server stores backups of configuration SQLite database files and intermediate files for configurations that were cleaned. Valid values: 0 through 24000 Default value: 30

Command Line Parameters for the Extractor

The following table describes a new parameter for the Extractor:

Command Line Parameter	New, Changed, or Deprecated	Description
SKIP_CONTINUITY_CHECK	Changed	This command-line parameter now allows to skip integrity checking for the database logs only during the first Extractor run.

Command Changes

Informatica Data Replication 9.7.0 introduces changes to the following commands:

- Server Manager CLI commands
- SQL Script Engine commands
- Tcl Script Engine commands

Server Manager CLI Commands

The following table describes changes to commands in the Server Manager Command Line Interface (CLI) in Data Replication 9.7.0:

Command	New, Changed, or Deprecated	Description
add connection	New	Creates a connection to a source or target. You must enter a unique connection name up to 120 characters in length.
delete connection	New	Deletes a connection to a source or target.
edit connection	New	Edits a connection to a source or target. The updated connection information will be used for all configurations to which the connection is assigned.
show connection info	Changed	Removed the optional id parameter for identifying a source or target connection by an internal ID number. You now identify a connection by entering the user-defined connection name only. Removed the optional --show-password parameter for displaying the encrypted password that the connection uses. For consistency with the Data Replication Console, encrypted connection passwords are hidden.

For more information, see the *Informatica Data Replication Command Line Interface for the Server Manager*.

SQL Script Engine Commands

The following table describes changes to the SQL Script Engine commands in Data Replication 9.7.0:

Command	New, Changed, or Deprecated	Description
:META.TRANS.USER_ID	Changed	This command now applies to DB2 for Linux, UNIX, and Windows and Microsoft SQL Server source types in addition to the previously supported Oracle source type.
:META.TRANS.USER_NAME	Changed	This command now applies to DB2 for Linux, UNIX, and Windows and Microsoft SQL Server source types in addition to the previously supported Oracle source type.
:NEW.*<column_name>*.type	Changed	This command now applies to Applier processing in Audit Apply and Merge Apply modes in addition to the previously supported Applier processing in SQL Apply mode and InitialSync processing.
:OLD.*<column_name>*.type	Changed	This command now applies to Applier processing in Audit Apply and Merge Apply modes in addition to the previously supported Applier processing in SQL Apply mode.

Tcl Script Engine Commands

The following table describes changes to the Tcl Script Engine commands in Data Replication 9.7.0:

Command	New, Changed, or Deprecated	Description
::dbs::transaction::user_id	Changed	This command now applies to DB2 for Linux, UNIX, and Windows and Microsoft SQL Server source types in addition to the previously supported Oracle source type.
::dbs::transaction::user_name	Changed	This command now applies to DB2 for Linux, UNIX, and Windows and Microsoft SQL Server source types in addition to the previously supported Oracle source type.

Changes to Supported Operating Systems and Databases

Data Replication 9.7.0 introduces the following changes to supported versions of databases:

Replication Sources

The following table describes changes to the replication sources that Data Replication supports:

Source	Versions	New or Deprecated
Amazon Relational Database Service (RDS) for Oracle	- 11*g* R2 - 12*c* R1	New
Microsoft SQL Server	- 2016	New
MySQL Enterprise Edition[1]	- 5.7.*x*	New

1. Data Replication 9.7.0 adds support for MySQL sources. You can run the Extractor for MySQL sources on Linux or Windows operating systems only. Version 5.7.*x* is supported for both MySQL sources and targets.

Replication Targets

The following table describes changes to the replication targets that Data Replication supports:

Target	Versions	New or Deprecated
Apache Kafka[1]	- 0.8.2.2 and later from any available binary download. The Scala version is immaterial. [2]	New
Cloudera	- 5.3	Deprecated
MySQL Enterprise Edition	- 5.5.*x*	Deprecated
MySQL Enterprise Edition	- 5.7.*x*	New

1. The Applier for Apache Kafka targets is supported on Linux and Windows operating systems only.
2. Data Replication does not use Scala binary code but does require the Kafka libraries from the binary download to be present.

CHAPTER 5

Version 9.6.4 HotFix 3

This chapter includes the following topic:

- New Features and Enhancements, 39

New Features and Enhancements

Informatica Data Replication 9.6.4 HotFix 3 does not include new features and enhancements. This hotfix provides the fixes that are described in the *Informatica 9.6.4 HotFix 3 Release Notes*.

CHAPTER 6

Version 9.6.4 HotFix 2

This chapter includes the following topic:

- New Features and Enhancements, 40

New Features and Enhancements

Informatica Data Replication 9.6.4 HotFix 2 does not include new features and enhancements. This hotfix provides the fixes that are described in the *Informatica 9.6.4 HotFix 2 Release Notes*.

CHAPTER 7

Version 9.6.4 HotFix 1

This chapter includes the following topics:

- New Features and Enhancements, 41
- Changes to Supported Operating Systems and Databases, 42

New Features and Enhancements

Informatica Data Replication 9.6.4 HotFix 1 includes the following new features and enhancements:

Reinstated Support for PostgreSQL Targets

Data Replication 9.6.4 HotFix 1 reinstates support for PostgreSQL targets. Supported PostgreSQL versions are 9.4, 9.5, and 9.6. Support for the PostgreSQL targets was temporarily suspended in Data Replication 9.6.1.

The following table describes changes to datatype mappings from Microsoft SQL Server sources to PostgreSQL targets:

Microsoft SQL Server Source Datatype	PostgreSQL Target Datatype	Comments

VARCHAR(*n*)	VARCHAR(*n*)	Prior to Data Replication 9.6.4 HotFix 1, the following PostgreSQL target datatype mappings were recommended for SQL Server source VARCHAR columns:
		- VARCHAR(*n*)
		Mapping condition:
		`n <= 255`
		- TEXT
		Mapping condition:
		`n > 255`

MONEY	NUMERIC(19,4)	Prior to Data Replication 9.6.4 HotFix 1, the recommended datatype mapping for SQL Server source MONEY columns was PostgreSQL target MONEY columns.

Changes to Supported Operating Systems and Databases

Data Replication 9.6.4 HotFix 1 introduces the following changes to supported versions of databases:

Replication Targets

The following table describes changes to the replication targets that Data Replication supports:

Target	Versions	New or Deprecated
PostgreSQL	- 9.4 - 9.5 - 9.6	New

CHAPTER 8

Version 9.6.4

This chapter includes the following topics:

- New Features and Enhancements, 43
- Parameter and Option Changes, 45
- Command Changes, 49
- Changes to Supported Operating Systems and Databases, 50

New Features and Enhancements

Informatica Data Replication 9.6.4 includes the following new features and enhancements:

Enhanced Security

Informatica Data Replication 9.6.4 includes the following new security features and enhancements:

Extended Session ID

After the Data Replication Console connects to the Server Manager Main server, the Server Manager assigns a session ID to the Console. All subsequent requests from the Data Replication Console to the Server Manager must include the session ID to authenticate the Console.

Previously, the Server Manager generated 4-byte session IDs in clear text. Now, the Server Manager generates 64-byte session IDs that are encoded with the Base64 algorithm for enhanced security.

Account Lockout

You can configure the maximum number of times that users can retry entering a password for logging in to the Server Manager Main server from the Data Replication Console or the Server Manager Command-Line Interface (CLI). If a user fails to enter a correct password after the maximum number of retries, the Server Manager temporarily locks out the user account.

Use the Server Manager MaxLoginAttempts and LoginBlockTimeout advanced properties to specify the maximum number of invalid login attempts and the account lockout duration.

Note: The idradmin user can unlock an account. To unlock an account, click **File** > **Users** and then click **Unlock** in the **Users** dialog box.

Password Strength

The Data Replication Console and the Server Manager CLI can now enforce the following password-strength policies when you create or edit a user account or connect to the Server Manager Main server:

- **Minimum password length**. Use the Server Manager CheckPasswordLength advanced property to specify the minimum password length.
- **Use of both uppercase and lowercase characters**. Use the Server Manager PasswordCaseSensitive advanced property to indicate whether passwords for user accounts must include both uppercase and lowercase characters.
- **Use of numbers**. Use the Server Manager PasswordContainNumbers advanced property to indicate whether passwords for user accounts must include one or more numbers.
- **Use of punctuation and symbols**. Use the Server Manager PasswordContainSpecialCharacters advanced property to indicate whether passwords for user accounts must include one or more special characters.

Password Expiration

Data Replication can now enforce a password expiration policy for user accounts. Use the Server Manager ExpirationPeriodForPasswords advanced property to specify the number of days after which passwords for user accounts expire. By default, passwords never expire.

Password for the idradmin Account

By default, the idradmin user has no password. If Data Replication is configured to enforce password-strength policies, the second time you connect to the Server Manager as the idradmin user, the Data Replication Console now requires you to specify a password for the idradmin account.

If you forget the password for the idradmin account, you can reset it to the default empty value. To reset the password, start the Server Manager Main server with the new reset_idradmin command-line parameter.

Secure Storage of Database Passwords

Data Replication can now encrypt the database passwords that you specify for source and target database connections by using the AES256 encryption algorithm.

Important: The Data Replication Console requires the Java Cryptography Extension (JCE) to handle encrypted database passwords.

Data Replication encrypts the database passwords with an internal master key. The Server Manager Main server generates this internal master key if you set the new Server Manager DBPasswordEncryptionEnabled advanced property to 1. Use the Server Manager Internal Master Key Path advanced property to define the path to the internal master key.

Data Replication uses the internal master key to decrypt the passwords that are required to connect to the source and target databases.

Managing Change Data Capture (CDC) for Remote Microsoft SQL Server Sources

Prior to Data Replication 9.6.4, the Server Manager could manage SQL Server CDC settings only if the SQL Server instance and the Server Manager ran on the same system. In Data Replication 9.6.4, the Server Manager can also manage CDC settings for remote Microsoft SQL Server sources.

If you plan to run the Server Manager on a computer that is remote from the SQL Server source, set the **remote admin connections** option of the SQL Server sp_configure stored procedure to 1. This setting enables the remote Server Manager to use a dedicated administrator connection (DAC) to the SQL Server source.

InitialSync Enhancements for Loading Data with the SQL Server Bulk Copy Program

The following enhancements have been made to InitialSync loading of data to Microsoft SQL Server targets with the SQL Server Bulk Copy Program (BCP):

- InitialSync, when used with BCP, can now process any Tcl scripts or SQL expressions that are defined for source tables that are mapped to Microsoft SQL Server target tables.

  Important: To process SQL expressions, InitialSync requires audit log tables. InitialSync uses the audit log tables to process the SQL expressions in SQL Apply mode.

  Prior to Data Replication 9.6.4, InitialSync switched to the ODBC driver to load data to tables that contain columns associated with Tcl scripts or SQL expressions.
- InitialSync, when used with BCP, can now use the International Components for Unicode (ICU) library to convert character data from the source database encoding to the target database encoding.
  Prior to Data Replication 9.6.4, InitialSync could use BCP to load only UTF-8 encoded data.
- InitialSync, when used with BCP, can now handle numeric data overflows for Oracle source NUMBER columns that are replicated to SQL Server target columns that have a compatible datatype.
  Prior to Data Replication 9.6.4, InitialSync could handle numeric data overflows for Microsoft SQL Server targets only when using the ODBC driver.

Oracle User Name Field in Audit Log Tables

Audit log tables for configurations with Oracle sources can now include a column that contains the name of the Oracle user who performed a DML operation. Data Replication includes this column in audit log tables only if the column name is specified in the **User name** field on the **Runtime Settings** tab > **Calculated Columns** view.

You can use the following SQL expression or Tcl command to get the user name value:

- :META.TRANS.USER_NAME in SQL expressions
- ::dbs::transaction::user_name in Tcl scripts

For more information, see the *Informatica Data Replication User Guide* and *Informatica Data Replication Scripting Guide*.

Secure Connections to Amazon Redshift Targets

Data Replication 9.6.4 can now use the TCP/IP protocol with SSL to securely connect to Amazon Redshift targets.

For more information, see the *Informatica Data Replication User Guide*.

Parameter and Option Changes

Informatica Data Replication 9.6.4 introduces changes to the following parameters and options:

- Advanced runtime parameters that you configure on the **Runtime Settings** tab > **Advanced Settings** view in the Data Replication Console.
- Server Manager advanced properties that you configure on the **Advanced Settings** tab of the **Properties** window in the Data Replication Console.

- Server Manager command-line parameters.

Advanced Runtime Parameters

The following table describes changes to advanced runtime parameters in Data Replication 9.6.4:

Advanced Runtime Parameter	New, Changed, or Deprecated	Description
extract.db2.readlog_input_endian_fix	Deprecated	This parameter is deprecated and is removed from the Data Replication Console for configurations with DB2 sources. If the Extractor task extracts data from a remote DB2 database server that has an architecture with a different endianness than the Extractor system, Data Replication now always converts the endianness for Extractor requests.

Advanced Properties for the Server Manager

The following table describes changes to Server Manager advanced properties in Data Replication 9.6.4:

Property	New, Changed, or Deprecated	Description
CheckPasswordLength	New	Specifies the minimum number of characters that passwords for user accounts must contain. If this property is set to 0, the Data Replication Console and the Server Manager Command-Line Interface (CLI) do not enforce a minimum password length. Default value: 0
DBPasswordEncryptionEnabled	New	Indicates whether Data Replication encrypts the database passwords that you specify for the source and target database connections. Options are: - **0**. Do not encrypt database passwords. - **1**. Encrypt database passwords by using the AES256 encryption algorithm. **Important:** If you did not add Java Cryptography Extension (JCE) files to the *JRE_HOME*/bin/security folder, the Data Replication Console does not allow you to set this property to 1. Default value: 0
ExpirationPeriodForPasswords	New	Specifies the number of days after which passwords for user accounts expire. When a password for a user account expires, the Server Manager Main server stops responding to user requests from the Data Replication Console. After the Server Manager Main server refuses a user request, the Data Replication Console or the Server Manager CLI in interactive mode prompts the user to change the expired password. Maximum value is 24000. To disable password expiration, set this property to 0. Default value: 0

Property	New, Changed, or Deprecated	Description
LoginBlockTimeout	New	Specifies the duration, in seconds, that a user account is locked out from logging in to the Server Manager Main server after the maximum number of failed login attempts that is specified in the MaxLoginAttempts property is reached. **Note:** The idradmin user can unlock an account. To unlock an account, click **File** > **Users** and then click **Unlock Account** in the **Users** dialog box. Default value: 1800 seconds (30 minutes)
MaxLoginAttempts	New	Specifies the maximum number of times that users can try to log in to the Server Manager Main server from the Data Replication Console or the Server Manager CLI. If a user fails to enter a correct password after the specified number of retries, the Server Manager locks out the user account for the duration that is specified in the LoginBlockTimeout property. To allow users an unlimited number of login attempts, set this property to 0. Default value: 5
PasswordCaseSensitive	New	Indicates whether passwords for user accounts must include both lowercase and uppercase alphabetic characters. Options are: - **0**. Passwords are not required to include both lowercase and uppercase alphabetic characters. - **1**. Passwords must include both lowercase and uppercase alphabetic characters. Default value: 0

Property	New, Changed, or Deprecated	Description
PasswordContainNumbers	New	Indicates whether passwords for user accounts must include one or more numbers. Options are: - **0**. Passwords are not required to include numbers. - **1**. Passwords must include numbers. Default value: 0
PasswordContainSpecialCharacters	New	Indicates whether passwords for user accounts must include one or more special characters. Options are: - **0**. Passwords are not required to include special characters. - **1**. Passwords must include at least one of the following special characters: - Space ( ) - Exclamation mark (!) - Double quotation mark (") - Hash sign (#) - Dollar sign ($) - Percent sign (%) - Ampersand (&) - Single quotation mark (') - Left parenthesis (() - Right parenthesis ()) - Asterisk (*) - Plus sign (+) - Comma (,) - Minus sign (-) - Full stop (.) - Slash (/) - Colon (:) - Semicolon (;) - Less than (<) - Equal sign (=) - Greater than (>) - Question mark (?) - At sign (@) - Left bracket ([) - Backslash (\) - Right bracket (]) - Caret (^) - Underscore (_) - Grave accent (`) - Left brace ({) - Vertical bar (|) - Right brace (}) - Tilde (~) Default value: 0

Command-Line Parameters for the Server Manager

The following table describes changes to Server Manager command-line parameters in Data Replication 9.6.4:

Property	New, Changed, or Deprecated	Description
reset_idradmin	New	Resets the password for the idradmin account to empty. This parameter also unlocks the idradmin account if the account is locked out after the maximum number of failed login attempts that is specified in the MaxLoginAttempts property is reached. **Note:** Do not use reset_idradmin in combination with other parameters.

Command Changes

Informatica Data Replication 9.6.4 introduces changes to the following commands:

- SQL Script Engine commands
- Tcl Script Engine commands

SQL Script Engine Commands

The following table describes changes to the SQL Script Engine commands in Data Replication 9.6.4:

Command	New, Changed, or Deprecated	Description
:META.TRANS.USER_NAME	New	For Applier processing of change data from Oracle sources, returns the name of the Oracle database user who performed a DML operation. **Note:** If the Oracle redo logs do not contain the user name, returns a null value. For Applier processing of change data from sources other than Oracle, returns a null value. For InitialSync processing, returns a null value.

Tcl Script Engine Commands

The following table describes changes to the Tcl Script Engine commands in Data Replication 9.6.4:

Command	New, Changed, or Deprecated	Description
::dbs::transaction::user_name	New	For Applier processing of change data from Oracle sources, returns the name of the Oracle database user who performed a DML operation. **Note:** If the Oracle redo logs do not contain the user name, returns a null value. For Applier processing of change data from sources other than Oracle, returns a null value. For InitialSync processing, returns a null value.

Changes to Supported Operating Systems and Databases

Data Replication 9.6.4 introduces the following changes to supported versions of operating systems and databases:

Replication Sources

The following table describes changes to the replication sources that Data Replication supports:

Source	Versions	New or Deprecated
Oracle	11*g* R1	Deprecated

Replication Targets

The following table describes changes to the replication targets that Data Replication supports:

Target	Versions	New or Deprecated
Cloudera	- 5.6 - 5.7	New
Microsoft SQL Server	SQL Server 2016 - targets only	New
Oracle	11*g* R1	Deprecated

CHAPTER 9

Version 9.6.3 HotFix 2

This chapter includes the following topics:

- New Features and Enhancements, 51
- Command Changes, 51

New Features and Enhancements

Informatica Data Replication 9.6.3 HotFix 2 includes the following new features and enhancements:

Oracle User Name Field in Audit Log Tables

Audit log tables for configurations with Oracle sources can now include a column that contains the name of the Oracle user who performed a DML operation. Data Replication includes this column in audit log tables only if the column name is specified in the **User name** field on the **Runtime Settings** tab > **Calculated Columns** view.

You can use the following SQL expression or Tcl command to get the user name value:

- :META.TRANS.USER_NAME in SQL expressions
- ::dbs::transaction::user_name in Tcl scripts

For more information, see the *Informatica Data Replication User Guide* and *Informatica Data Replication Scripting Guide*.

Command Changes

Informatica Data Replication 9.6.3 HotFix 2 introduces changes to the following commands:

- SQL Script Engine commands
- Tcl Script Engine commands

SQL Script Engine Commands

The following table describes changes to the SQL Script Engine commands in Data Replication 9.6.3 HotFix 2:

Command	New, Changed, or Deprecated	Description
:META.TRANS.USER_NAME	New	For Applier processing of change data from Oracle sources, returns the name of the Oracle database user who performed a DML operation. **Note:** If the Oracle redo logs do not contain the user name, returns a null value. For Applier processing of change data from sources other than Oracle, returns a null value. For InitialSync processing, returns a null value.

Tcl Script Engine Commands

The following table describes changes to the Tcl Script Engine commands in Data Replication 9.6.3 HotFix 2:

Command	New, Changed, or Deprecated	Description
::dbs::transaction::user_name	New	For Applier processing of change data from Oracle sources, returns the name of the Oracle database user who performed a DML operation. **Note:** If the Oracle redo logs do not contain the user name, returns a null value. For Applier processing of change data from sources other than Oracle, returns a null value. For InitialSync processing, returns a null value.

CHAPTER 10

Version 9.6.3 HotFix 1

This chapter includes the following topic:

- New Features and Enhancements, 53

New Features and Enhancements

Informatica Data Replication 9.6.3 HotFix 1 does not include new features and enhancements. This hotfix provides the fixes that are described in the *Informatica 9.6.3 HotFix 1 Release Notes*.

CHAPTER 11

Version 9.6.3

This chapter includes the following topics:

- New Features and Enhancements, 54
- Parameter and Option Changes, 55
- Command Changes, 57
- Changes to Supported Operating Systems and Databases, 59

New Features and Enhancements

Informatica Data Replication 9.6.3 includes the following new features and enhancements:

Enhanced Support for Oracle 12c Sources

Data Replication 9.6.2 introduced the ability to capture change data from Oracle 12*c* sources but did not support any of the features that Oracle 12*c* introduced. Data Replication 9.6.3 now replicates data from sources that use the following Oracle 12*c* features:

- Columns that have extended datatypes of up to 32,767 bytes in size, including NVARCHAR2, RAW, and VARCHAR2 columns.
- Tables for which Oracle In-Database Archiving is enabled. However, Data Replication does not replicate the ROW ARCHIVAL clause and does not replicate DML or DDL operations for the hidden system column ORA_ARCHIVE_STATE.
- Columns that are defined as invisible.
- Numeric columns that are defined with the IDENTITY clause.
  If a source DDL statement contains the IDENTITY clause, the Applier executes the equivalent DDL statement on the target but without the IDENTITY clause.
- Databases that use multitenant architecture. For Data Replication to capture changes, the source tables must reside in a single pluggable database (PDB) within a multitenant container database (CDB). In the Data Replication Console, enter the name of the PDB in the **Instance** field on the **Source Database** tab.

For more information, see the *Informatica Data Replication User Guide* and the *Informatica Data Replication Datatype Mapping Reference*.

Dynamic Allocation of Server Manager Threads

Prior to Data Replication 9.6.3, the Server Manager could not dynamically allocate threads for running replication tasks, processing incoming requests, and sending messages to other Server Manager instances. Instead, you specified the maximum number of active threads for these tasks in the following Server Manager properties:

- ApiThreadsCount
- BuiltinTaskManagersCountThreads
- IntermediateFileTransfersCountThreads
- SendCountThreads

In Data Replication 9.6.3, the Server Manager allocates threads for these tasks dynamically by default. To disable dynamic allocation of threads in a Server Manager instance, set the DynamicThreadAllocation property to 0.

For more information, see the *Informatica Data Replication User Guide*.

Oracle User ID Information

For Oracle sources, audit log tables can now include a column that contains the user ID of the Oracle user who performed a DML operation.

Data Replication includes this column in audit log tables only if the column name is specified in the **User ID** field on the **Runtime Settings** tab > **Calculated Columns** view.

You can use the following SQL expression or Tcl command to get the user ID value:

- :META.TRANS.USER_ID in SQL expressions
- ::dbs::transaction::user_id in Tcl scripts

Parameter and Option Changes

Informatica Data Replication 9.6.3 introduces changes to the following parameters and options:

- Advanced runtime parameters that you configure on the **Runtime Settings** tab > **Advanced Settings** view in the Data Replication Console.
- Advanced Server Manager properties that you configure on the **Advanced Settings** tab of the **Properties** window in the Data Replication Console.

Advanced Runtime Parameters

The following table describes changes to advanced runtime parameters in Data Replication 9.6.3:

Advanced Runtime Parameter	New, Changed, or Deprecated	Description
extract.mssql.process_updates_as_updates	Changed	For Microsoft SQL Server 2012 and later sources, if you set this parameter to 0, the Extractor now splits each Update operation into a pair of Delete and Insert operations.
general.number_overflow_abort	Changed	This parameter now applies to DB2 for Linux, UNIX, and Windows and Microsoft SQL Server target types in addition to the following previously supported target types: - Amazon Redshift - Greenplum - Netezza - Teradata - Vertica
general.number_overflow_replace	Changed	This parameter now applies to DB2 for Linux, UNIX, and Windows and Microsoft SQL Server target types in addition to the following previously supported target types: - Amazon Redshift - Greenplum - Netezza - Teradata - Vertica
general.number_overflow_verify	Changed	This parameter now applies to DB2 for Linux, UNIX, and Windows and Microsoft SQL Server target types in addition to the following previously supported target types: - Amazon Redshift - Greenplum - Netezza - Teradata - Vertica

Advanced Properties for the Server Manager

The following table describes changes to Server Manager advanced properties in Data Replication 9.6.3:

Property	New, Changed, or Deprecated	Description
DynamicThreadAllocation	New	Determines whether the Server Manager dynamically allocates threads for running replication tasks, processing incoming requests, and sending messages to other Server Manager instances. Options are: - **0**. Do not allocate Server Manager threads dynamically. Use the ApiThreadsCount, BuiltinTaskManagersCountThreads, IntermediateFileTransfersCountThreads, and SendCountThreads properties to specify the number of active threads that the Server Manager allocates to these tasks. - **1**. Allocate Server Manager threads dynamically. The Server Manager uses the values of the ApiThreadsCount, BuiltinTaskManagersCountThreads, IntermediateFileTransfersCountThreads, and SendCountThreads properties for initial allocation of threads. If necessary, the Server Manager can create more threads and then close threads after they complete their tasks. Default is 1.
HttpKeepAliveTimeout	Changed	The default value changed from 1000 to 10000.

Command Changes

Informatica Data Replication 9.6.3 introduces changes to the following commands:

- Server Manager CLI commands
- SQL Script Engine commands
- Tcl Script Engine commands

Server Manager CLI Commands

The following table describes changes to commands in the Server Manager Command Line Interface (CLI) in Data Replication 9.6.3:

Command	New, Changed, or Deprecated	Description
ping	Changed	Includes the following new, optional parameters that you can use to verify user credentials when you ping the Server Manager Main server: - login. A user name that has the authority to connect to the specified Server Manager Main server instance. - passwd. A valid password for the specified login user. Includes the following new, optional parameters that you can use to ping the Server Manager Main server or a subserver when using an HTTPS connection: - pathToCert. If the protocol parameter is set to HTTPS, the full path to the cert.pem certificate file. - pathToTruststore. If the protocol parameter is set to HTTPS, the full path to the TrustStore file that Java uses to store trusted certificates. - storepass. If the protocol parameter is set to HTTPS, the password for the Java TrustStore.
supplog run	Changed	Includes the following new optional parameters: - cdc_disable. Indicates how to handle the CDC capture and cleanup jobs that Microsoft SQL Server generates for databases with Change Data Capture enabled. - restart_instance. Indicates whether the Server Manager restarts the SQL Server Enterprise Edition instance in single-user mode to enable a dedicated administrator connection to the source database when you edit the Change Data Capture settings for source tables.

SQL Script Engine Commands

The following table describes changes to the SQL Script Engine commands in Data Replication 9.6.3:

Command	New, Changed, or Deprecated	Description
:META.TRANS.USER_ID	New	For Applier processing of change data from Oracle sources, returns the user ID of the Oracle database user who performed a DML operation. For Applier processing of change data from sources other than Oracle, returns a null value. For InitialSync processing, returns a null value

Tcl Script Engine Commands

The following table describes changes to the Tcl Script Engine commands in Data Replication 9.6.3:

Command	New, Changed, or Deprecated	Description
::dbs::transaction::user_id	New	For Applier processing of change data from Oracle sources, returns the user ID of the Oracle database user who performed a DML operation. For Applier processing of change data from sources other than Oracle, returns a null value. For InitialSync processing, returns a null value.

Changes to Supported Operating Systems and Databases

Data Replication 9.6.3 introduces the following changes to supported versions of operating systems and databases:

Replication Targets

The following table describes changes to the replication targets that Data Replication supports:

Target	Versions	New or Deprecated
Cloudera	5.2	Deprecated
Cloudera	- 5.4 - 5.5	New

CHAPTER 12

Version 9.6.2

This chapter includes the following topics:

- New Features and Enhancements, 60
- Behavior Changes, 62
- Parameter and Option Changes, 63
- Command Changes, 71
- Changes to Supported Operating Systems and Databases, 73

New Features and Enhancements

Informatica Data Replication 9.6.2 includes the following new features and enhancements:

Improved Interface for Deploying Configurations

In the Data Replication Console, the new **Deploy a Configuration** wizard guides you through configuring a full or partial deployment of a replication configuration locally or to a remote environment. The wizard is designed with usability features such as tooltips and a command button to test a connection to an Oracle ASM instance.

Also, the deployment feature provides the following new files to help you configure and monitor deployments:

- A sample deploy.properties file in the *DataReplication_installation*/doc/samples/deploy subdirectory, which you can use to configure a deployment
- A deployment.log file in the *DataReplication_installation*/logs subdirectory, which contains the error, warning, and informational messages that were issued during deployment of a configuration from the Data Replication Console

For more information, see the "Managing Replication Configurations" chapter in the *Informatica Data Replication User Guide*.

Reinstated Sources and Targets

The Data Replication 9.6.2 release reinstates support for the following sources and targets that were temporarily not supported in version 9.6.1:

- Microsoft SQL Server sources.
- Cloudera, DB2 for Linux, UNIX, and Windows, flat files, Hortonworks, MySQL, and Vertica targets.

The Data Replication 9.6.2 documentation covers only the source and target combinations that are supported in the Data Replication 9.6.2 release.

Support for Amazon Redshift Targets

Data Replication can now replicate source data to Amazon Redshift target tables.

Data Replication connects to and writes source data to the Amazon Simple Storage Service (Amazon S3). Amazon S3 is a storage service that can copy data from a source and simultaneously move data to Amazon Redshift clusters. After the source data is in Amazon S3 storage, Data Replication issues a copy command that copies the data to the Amazon Redshift target table.

Important: Data Replication cannot replicate data from source columns that have binary datatypes to Amazon Redshift targets.

For more information, see the *Informatica Data Replication User Guide*.

Enhanced Support for Database Character Set Conversion

Data Replication now supports character set conversion for replication configurations that have a DB2 for Linux, UNIX, and Windows, Microsoft SQL Server, or Oracle source and an Amazon Redshift, Greenplum, Netezza, Teradata, or Vertica target.

Important: For other types of targets, ensure that the source and target databases use the UTF-8 or UTF-16 encoding for character data.

For more information, see the *Informatica Data Replication User Guide*.

Enhanced Support for Managing Database Supplemental Logging

Data Replication 9.6.2 now provides the ability to manage database supplemental logging for both mapped and unmapped tables in all types of source databases.

- For Oracle sources, Data Replication 9.6.1 provided the ability to view, add, and remove columns in supplemental log groups for mapped tables only. Data Replication 9.6.2 extends support to unmapped tables. You can manage columns in supplemental log groups for mapped and unmapped tables that are associated with a configuration from the **Map Tables** tab in the Data Replication Console or with the Server Manager CLI supplog commands.
- For Microsoft SQL Server sources, most Data Replication releases earlier than 9.6.1 provided the ability to manage Replicate mode settings for SQL Server source tables from the Data Replication Console or Server Manager CLI commands. These settings were stored in the Server Manager SQLite database. In Data Replication 9.6.2, for consistency with the currently supported SQL Server versions, you now manage SQL Server Change Data Capture settings for the source database and its source tables, instead of setting Replicate mode. You can enable or disable SQL Server Change Data Capture for mapped and unmapped source tables that are associated with a configuration from the **Map Tables** tab in the Data Replication Console or by using the Server Manager CLI supplog commands.
- For DB for Linux, UNIX, and Windows sources, Data Replication 9.6.2 introduces support for managing database supplemental logging. You can set the DB2 DATA CAPTURE CHANGES or DATA CAPTURE NONE option for mapped or unmapped source tables that are associated with a configuration. You can manage these settings from the **Map Tables** tab in the Data Replication Console or with the Server Manager CLI supplog commands.

For all source types, Data Replication generates SQL statements that implement the appropriate source database options. You can run the statements on demand, or if you do not have the authority to run the statements, you can save them to a script file and give the file to your database administrator.

For more information, see the "Managing Replication Configurations" chapter in the *Informatica Data Replication User Guide* and the "Command Reference" chapter in the *Informatica Data Replication Command Line Interface for the Server Manager.*

Behavior Changes

Data Replication 9.6.2 introduces the following behavior changes.

Disabling Microsoft SQL Server CDC Jobs

Prior to Data Replication 9.6.2, the Server Manager did not disable the Microsoft SQL Server CDC capture and cleanup jobs after enabling Change Data Capture for Microsoft SQL Server source databases. If you did not run Microsoft SQL Server native replication and the Data Replication Extractor against the same database, you had to disable the CDC jobs manually.

In Data Replication 9.6.2, the Server Manager still does not disable the Microsoft SQL Server CDC jobs, by default, after the Server Manager enables SQL Server Change Data Capture. However, in the Data Replication Console, you can disable all SQL Server CDC jobs, regardless of whether SQL Server Change Data Capture is enabled or not enabled for the source tables. Set the Server Manager CDCJobsDisableMode property to **2**, or select **Always disable** in the **SQL Server CDC Jobs** list in the **Microsoft SQL Server Instances** dialog box.

Alternatively, if SQL Server Change Data Capture is not enabled for any of the source tables when you create a configuration, you can disable the SQL Server CDC jobs. Set the CDCJobsDisableMode property to **1**, or select **Disable only if CDC is not enabled** in the **SQL Server CDC Jobs** list in the **Microsoft SQL Server Instances** dialog box.

For more information, see the "Configuring the Server Manager" chapter in the *Informatica Data Replication User Guide*.

Changes to DB2 Source and Target Connectivity for Data Load Operations

Prior to Data Replication 9.6.2, InitialSync could use the DB2 API or ODBC drivers to unload data from a DB2 source or load data to a DB2 target.

In Data Replication 9.6.2, InitialSync can use only ODBC drivers for DB2 sources and targets.

Discontinued Support for Comparing Data in Mapped Source and Target Tables

Prior to Data Replication 9.6.2, the **Compare Tables** button on the **Map Tables** tab provided the ability to compare data in the mapped tables for a configuration.

In Data Replication 9.6.2, you can no longer compare data in mapped tables from the Data Replication Console.

Replicating Named Inline Constraints for Table Columns

Prior to Data Replication 9.6.2, Data Replication could not replicate named inline constraints for source table columns. If you replicated a CREATE TABLE operation that defined a named constraint on a source column, the Applier did not name the corresponding target constraint. To replicate a named constraint for a source column, you had to add the constraint in a separate DDL statement, for example:

```
CREATE TABLE schema.table (a INTEGER, b INTEGER);
ALTER TABLE schema.table
ADD CONSTRAINT constraint PRIMARY KEY (b);
```

Data Replication 9.6.2 can replicate named inline constraints for source table columns. When you replicate a CREATE TABLE operation that defines a named inline constraint on a source column, the Applier replicates

the constraint with the same constraint name to the target. For example, Data Replication can now replicate the following DDL operation:

```
CREATE TABLE schema.table (a INTEGER, b INTEGER CONSTRAINT constraint PRIMARY KEY);
```

For more information, see the "DDL Replication" chapter in the *Informatica Data Replication User Guide*.

Distribution by Tables for DB2 Targets

Prior to Data Replication 9.6.2, DB2 target tables used distribution by rows as the default algorithm for distributing Insert, Update, and Delete records across threads for parallel apply processing.

In Data Replication 9.6.2, DB2 targets use distribution by tables by default to minimize deadlocks and lock contention among threads. With this option, the change records for each table are assigned to a separate thread.

For more information, see the "Creating Replication Configurations in the Data Replication Console" chapter in the *Informatica Data Replication User Guide*.

Parameter and Option Changes

Informatica Data Replication 9.6.2 introduces changes to the following parameters and options:

- Advanced runtime parameters that you configure on the **Runtime Settings** tab > **Advanced Settings** view in the Data Replication Console.
- Advanced Server Manager properties that you configure on the **Advanced Settings** tab of the **Properties** window in the Data Replication Console.
- Command line parameters for the Extractor.

Advanced Runtime Parameters

The following table describes changes to advanced runtime parameters in Data Replication 9.6.2:

Advanced Runtime Parameter	New, Changed, or Deprecated	Description
apply.enable_greenplum_table_redistribution	New	Indicates whether the Applier runs an ALTER TABLE *table* SET DISTRIBUTED *column*,... statement before replicating a DDL operation that adds a primary key constraint or unique index to a Greenplum target. Options are: - **0**. Do not run the ALTER TABLE ... SET DISTRIBUTED statement. **Important:** If the target table uses the random distribution policy, the Applier cannot replicate DDL operations that add a primary key constraint or unique index. - **1**. Run the ALTER TABLE ... SET DISTRIBUTED statement on a Greenplum target before applying a DDL operation that adds a primary key constraint or unique index. This option sets the distribution key columns to the unique constraint columns. **Important:** If the target table contains a large number of rows, this option can be resource intensive. Default value: 1 Change the default value only at the request of Informatica Global Customer Support.
apply.loader_tx_map_size_tx	Changed	The default value changed from 32768 to 500000.
apply.trace_db_execution	Changed	The parameter options for enabling SQL traces have changed. Changed option: - **1**. Use standard SQL trace level 1. Previously, this option enabled the SQL trace level specified in the apply.trace_db_execution_waits parameter. New options: - **4**. Use SQL trace level 4. This option displays the level 1 information and additional information about bind variables. - **8**. Use SQL trace level 8. This option displays the level 1 information and additional information about wait time. - **12**. Use SQL trace level 12. This option enables trace levels 4 and 8 simultaneously. **Note:** Option **0** has not changed. It still disables SQL tracing.
apply.trace_db_execution_waits	Deprecated	Use the apply.trace_db_execution parameter.

Advanced Runtime Parameter	New, Changed, or Deprecated	Description
extract.mssql.updates_as_updates_lookup_size	Obsolete	This parameter is deprecated because the Extractor now uses a different procedure to merge the Delete and Insert operations that Microsoft SQL Server 2008 R2 creates for an Update into a single Update operation.
extract.oracle.check_size_of_parsed_logs	Changed	The default value is now 1.
extract.oracle.read_log_list_attempts	New	Specifies the maximum number of times the Extractor tries to get the list of Oracle redo log files for processing. If the redo logs do not pass integrity checking after the Extractor reaches the specified maximum number of attempts, the Extractor stops processing the redo logs. Default value: 5

extract.oracle.threads_offline_at_start	New	For Oracle RAC sources, specifies a comma-separated list of the Oracle redo threads that are offline at the SCN that corresponds to the Start SCN value for the Extractor. The Extractor starts processing change data records from an Oracle redo thread that is in this list only after a Thread Enable Marker (TEM) appears in the redo log. The Data Replication Console populates this list of offline redo threads when you save the replication configuration. You can also specify the list of offline redo threads in this parameter manually. Use a comma (,) as a separator. To determine the redo threads that are offline at a specific Start SCN point, run the following statement:
		`SELECT thread# FROM v$thread`
		`MINUS`
		`  SELECT thread# FROM v$archived_log`
		`  WHERE first_change# <= Start_SCN AND next_change# > Start_SCN`
		`GROUP BY thread#;`

Advanced Runtime Parameter	New, Changed, or Deprecated	Description
flat_file.save_to_cycle1_dir	New	For Cloudera, Flat File, and Hortonworks targets, specifies the naming pattern that the Applier uses to create output directories for replication files. Replication files are .txt files that contain change data. Options are: - **0**. For each apply cycle, the Applier creates the cycle_*cycle_sequence_number*.work subdirectory that contains replication files. - **1**. For each apply cycle, the Applier creates the cycle_1.work.*sequence_number* subdirectory that contains replication files. During the last cycle, Data Replication stores replication files in the cycle_1.work subdirectory. When you run replication jobs later, Data Replication appends a sequence number to the directory name. **Note:** Use this option for backward compatibility with earlier Data Replication versions that used that directory naming pattern. Default value: 0
general.number_overflow_abort	Changed	This parameter additionally applies to Amazon Redshift and Vertica target types. All of the following target types are now supported: - Amazon Redshift - Greenplum - Netezza - Teradata - Vertica
general.number_overflow_replace	Changed	This parameter additionally applies to Amazon Redshift and Vertica target types. All of the following target types are now supported: - Amazon Redshift - Greenplum - Netezza - Teradata - Vertica
general.number_overflow_verify	Changed	This parameter additionally applies to Amazon Redshift and Vertica target types. All of the following target types are now supported: - Amazon Redshift - Greenplum - Netezza - Teradata - Vertica

Advanced Runtime Parameter	New, Changed, or Deprecated	Description
global.fix_invalid_db_characters	New	For Amazon Redshift, Greenplum, and Teradata targets, indicates whether to fix UTF-8 encoded characters in character data that the targets cannot process correctly during InitialSync and Applier processing. **Note:** The UTF-8 encoded characters are valid but the targets do not support them. Options are: - **0**. Do not fix UTF-8 encoded symbols in the character data that the target cannot process correctly. The InitialSync and Applier tasks might fail. - **1**. Replace each byte of UTF-8 encoded symbols in character data that the target cannot process correctly with the character that is specified in global.invalid_character_replacement parameter. Default value: 0
global.fix_invalid_encoding_characters	New	For Amazon Redshift, Greenplum, Netezza, Teradata, and Vertica targets, indicates whether to fix invalid UTF-8 byte sequences in character data during InitialSync and Applier processing. Options are: - **0**. Do not fix invalid UTF-8 byte sequences. The InitialSync and Applier tasks might fail. - **1**. If the global.icu_enabled parameter is set to 1 or the source data contains UTF-16 characters, replace invalid UTF-8 byte sequences with the character that is specified in the global.invalid_character_replacement parameter. If the global.icu_enabled parameter is set to 0 or the source data contains only UTF-8 characters, replace each byte of the invalid UTF-8 byte sequences with the character that is specified in the global.invalid_character_replacement parameter. Default value: 0
global.icu_enabled	Changed	The default value is now 0. The Data Replication Console sets this parameter to 1 for new configurations.

Advanced Runtime Parameter	New, Changed, or Deprecated	Description
global.invalid_character_replacement	New	Specifies the ASCII character that replaces UTF-8 encoded characters during InitialSync and Applier processing in the following situations: - If you set the global.fix_invalid _db_characters parameter to 1 for Greenplum and Teradata targets, replaces UTF-8 encoded characters that the targets cannot process correctly. - If you set the global.fix_invalid_encoding_characters parameter to 1 for Amazon Redshift, Greenplum, Netezza, Teradata, and Vertica targets, replaces invalid UTF-8 byte sequences. Default value: ? (question mark)
global.lsn_table_connection_attempts	New	For DB2 sources, indicates the number of times the Extractor or InitialSync retries a query to the DBSYNC_SYNC_LSN service table after the initial query timeout interval elapses. The query timeout interval is specified in the global.lsn_table_connection_timeout parameter. Default value: 3
global.lsn_table_connection_timeout	New	For DB2 sources, indicates the number of seconds after which an Extractor or InitialSync query to the DBSYNC_SYNC_LSN service table will time out. After this interval elapses, the Extractor or InitialSync retries the query up to the number of attempts specified in the global.lsn_table_connection_attempts parameter. Default value: 15
initial.db2.export_enabled	Obsolete	This parameter is obsolete and will be removed from the configuration. InitialSync now always uses ODBC drivers for the initial unload of data for DB2 sources.
initialsync.oracle.9i_flashback_ workaround_enabled	Deprecated	This parameter is deprecated because Oracle 9*i* sources and targets are no longer supported.
oracle.truncate_char_to_4000	New	For Oracle targets, indicates whether the Applier truncates character data to 4,000 bytes if the corresponding source tables have character data that is larger than 4,000 bytes. Options are: - **0**. Do not truncate character data. - **1**. Truncate character data to 4,000 bytes. Default value: 0
redshift.s3.bucket_name	New	Specifies the name of the Amazon S3 bucket that stores the temporary files with the data to be applied to the Amazon Redshift target.

Advanced Runtime Parameter	New, Changed, or Deprecated	Description
redshift.s3.file_size	New	Specifies the maximum size of the temporary files, in kilobytes, that Data Replication creates in the Amazon S3 bucket before loading the source data to Amazon Redshift. Default value: 5120 KB
redshift.s3.key_id	New	Specifies an AWS access key ID that Data Replication must use to access the Amazon S3 account resources including the bucket where Data Replication creates temporary files.
redshift.s3.path	New	Specifies the name of the directory in the Amazon S3 bucket where Data Replication creates the temporary files that store source data.
redshift.s3.secret_key	New	Specifies the secret access key for the access key ID that is specified in the redshift.s3.key_id parameter. The access key ID must have the authority to access the Amazon S3 account resources.
teradata.fix_invalid_utf8	Deprecated	Use the global.fix_invalid_encoding_characters runtime parameter.
teradata.fix_td_invalid_utf8	Deprecated	Use the global.fix_invalid_db_characters runtime parameter.
teradata.utf8_replace_character	Deprecated	Use the global.invalid_character_replacement runtime parameter.

Advanced Properties for the Server Manager

The following table describes changes to Server Manager advanced properties in Data Replication 9.6.2:

Property	New, Changed, or Deprecated	Description
CdcJobsDisableMode	New	For configurations with Microsoft SQL Server sources, determines whether the Server Manager disables the SQL Server CDC capture and cleanup jobs after enabling SQL Server Change Data Capture for the source database and tables. Options are: - **0**. Do not disable the SQL Server CDC jobs. Select this option if you do not want Data Replication to disable the CDC jobs because the jobs have already been disabled from Microsoft SQL Server Management Studio or you accept that Data Replication might not capture all of the change data. - **1**. Disable the SQL Server CDC jobs if SQL Server Change Data Capture is not enabled for any source tables when you create the configuration in the Data Replication Console. With this option, the SQL Server CDC jobs are not disabled if SQL Server Change Data Capture is enabled for any source tables when you create the configuration. - **2**. Disable the SQL Server CDC jobs regardless of whether SQL Server Change Data Capture is enabled or not enabled for the source tables. Default is 0. The options for this property are equivalent to the options for the **SQL Server CDC Jobs** field in the **Microsoft SQL Server Instances** dialog box. When you edit either this property or the **SQL Server CDC Jobs** setting, the other value is updated to the equivalent setting.
EnableMSSQLCDCJobCleanup	Deprecated	Use the new CdcJobsDisableMode parameter.
RestartMSSQLInstance	New	For configurations with Microsoft SQL Server Enterprise Edition sources, determines whether the Server Manager restarts the SQL Server instance in single-user mode to enable a dedicated administrator connection to the source database when you edit the Change Data Capture settings for source tables on the **Map Tables** tab. Options are: - **0**. The Server Manager does not restart the SQL Server instance when you change the Change Data Capture setting for a source table. - **1**. The Server Manager restarts the SQL Server instance every time you change the Change Data Capture setting for a source table. Default is 0. For Microsoft SQL Server Standard Edition sources, the Server Manager always restarts the SQL Server instance when you edit the Change Data Capture setting for a source table, regardless of the value for this property. The options for this property are equivalent to the **Restart instance** settings in the **Microsoft SQL Server Instances** dialog box. When you edit either this property or the **Restart instance** setting, the other value is updated to the equivalent setting.

Command Line Parameters for the Extractor

The following table describes a new parameter for the Extractor:

Command Line Parameter	New, Changed, or Deprecated	Components	Description
SKIP_TEM_ISSUES_FOR_THREADS= *redo_thread1,redo_thread2, ...*	New	Extractor	For Oracle RAC sources, specifies the list of redo threads for which the Extractor ignores Thread Enable Marker (TEM) issues. If a TEM issue occurs, the Extractor continues processing redo log records. The following types of TEM issues indicate that the Extractor might have missed change data records for a redo thread: - The Extractor encounters a TEM for an online Oracle redo thread. - A TEM for an offline redo thread refers to an unexpected log sequence number. Ignoring these TEM issues for a redo thread might result in change data loss. Use this parameter only if you must continue Extractor processing despite possible change data loss. Default value: None.

Command Changes

Informatica Data Replication 9.6.2 introduces changes to the following commands:

- Server Manager CLI commands

Server Manager CLI Commands

The following table describes changes to commands in the Server Manager Command Line Interface (CLI) in Data Replication 9.6.2:

Command	New, Changed, or Deprecated	Description
connect	Changed	Includes the following new, optional parameters to connect to the Server Manager Main server with the HTTPS protocol: - pathToCert. The full path to the cert.pem certificate file. - pathToTruststore. The full path to the TrustStore file that Java uses to store trusted certificates. - storepass. The password for the Java TrustStore.
redeploy config	Changed	Includes the new, optional -f parameter to stop all running tasks for the configuration in the destination environment.

Command	New, Changed, or Deprecated	Description
remove thread	Deprecated	The Extractor can now handle starting and stopping Oracle RAC nodes.
supplog add	Changed	Includes the new, optional -a parameter to enable database supplemental logging for an unmapped source table. - For a DB2 source, sets DATA CAPTURE CHANGES. - For a Microsoft SQL Server source, enables Change Data Capture. - For an Oracle source, adds source table columns to a supplemental log group. After the supplog add command runs successfully, the Server Manager CLI now lists the number of updated objects.
supplog refresh	New	Refreshes the database supplemental logging settings directly from the source database. For Microsoft SQL Server sources, also refreshes the Server Manager cache.
supplog remove	Changed	Includes the new, optional -a parameter to disable database supplemental logging for an unmapped source table. - For a DB2 source, sets DATA CAPTURE NONE. - For a Microsoft SQL Server source, disables Change Data Capture. - For an Oracle source, removes source table columns from a supplemental log group.
supplog run	Changed	Includes the new, optional -r parameter that allows the Server Manager to restart Microsoft SQL Server Standard Edition sources to change the Change Data Capture setting. Use this parameter in non-interactive mode only.
supplog show	Changed	Includes the following new, optional parameters: - -t. Shows only tables that do not have database supplemental logging enabled. - -a. Shows all mapped and unmapped tables for the specified source schema. - -c. For Oracle sources only, shows only the columns in the specified source tables that are not included in a supplemental log group that Data Replication created. The optional -allcolumns parameter for Oracle sources is deprecated.

Changes to Supported Operating Systems and Databases

Data Replication 9.6.2 introduces the following changes to supported versions of operating systems and databases:

Operating Systems

The following table describes changes to the operating system versions that Data Replication supports:

Operating System	Versions	New or Deprecated
Redhat Enterprise Linux	5.*x*	Deprecated
Redhat Enterprise Linux	7	New
Solaris	10	Deprecated
SUSE Linux Enterprise Server	12	New

Replication Sources

The following table describes changes to the replication sources that Data Replication supports:

Source	Versions	New or Deprecated
Microsoft SQL Server	- SQL Server 2005 - SQL Server 2008 R1	Deprecated
Microsoft SQL Server [1]	- SQL Server 2012 - SQL Server 2014	New
Oracle	- 12*c* [2]	New

1. Data Replication supports only Microsoft SQL Server Standard and Enterprise Edition sources.

2. Data Replication can capture change data from Oracle 12*c* sources but does not support any of the new features that Oracle 12*c* introduced. For example, Data Replication cannot extract change data from identity columns, columns that have extended datatypes, and columns that are defined as invisible.

Replication Targets

The following table describes changes to the replication targets that Data Replication supports:

Target	Versions	New or Deprecated
Amazon Redshift	-	New
Cloudera	- 5.2 - 5.3	New
DB2 for Linux, UNIX, and Windows	- 9.5	Deprecated
DB2 for Linux, UNIX, and Windows	- 10.1 - 10.5	New

Target	Versions	New or Deprecated
Hadoop Distributed File System (HDFS)	- Apache Hadoop 0.20.203.*x* - Apache Hadoop 0.20.204.*x* - Apache Hadoop 0.20.205.*x* - Apache Hadoop 0.23.*x* - Apache Hadoop 1.0.*x* - Apache Hadoop 1.1.*x* - Apache Hadoop 2.*x*.*x* - Cloudera CDH4 - Cloudera CDH3	Deprecated
Hortonworks	- HortonWorks 2.2 - HortonWorks 2.3	New
Microsoft SQL Server	- SQL Server 2008 [1]	Deprecated
Microsoft SQL Server	- SQL Server 2014	New
MySQL Enterprise Edition	- 5.1	Deprecated
Oracle	- 12*c*	New
Teradata	- 13.10 - 14.0	Deprecated
Teradata	- 15.10	New
Vertica	- 5.1	Deprecated
Vertica	- 7.*x*	New
1. Data Replication supports Microsoft SQL Server 2008 R2		

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